Review

Eigenvalue equation :

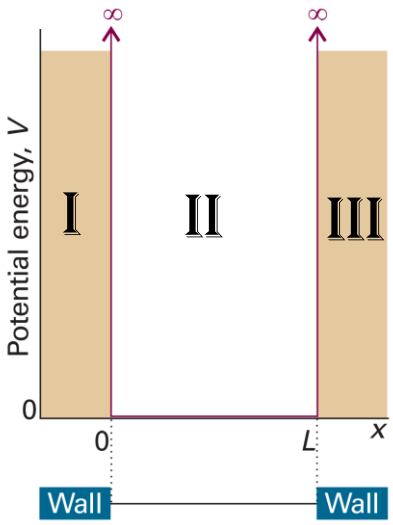
Eigenfunction
Eigenvalue

Two properties of the Hermitian operators:

1)

2)

Orthonormal



A particle in a 1-D box

$$V(x) = \infty \ , \ x \in \mathrm{I}, \mathrm{III} \ ,$$

$$V(x) = 0 \ , \ x \in \mathrm{II}$$

“Bound state”

“Unbound state”

Probability density:

Review

3-D one-particle Schrödinger equation

Laplacian Operator

Normalization condition:

The particle in a 3-D box

$V(x,y,z)=0 \quad 0<x<a,\ \ 0<y<b,\ \ 0<z<c$

$V(x,y,z)=\infty \quad$ *elsewhere*

Degenerate **Degree of degeneracy**

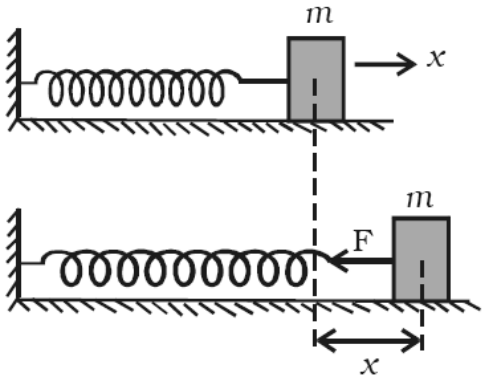
$\hat{H}\psi_1 = w\psi_1,\ \ \hat{H}\psi_2 = w\psi_2,\cdots,\ \ \hat{H}\psi_n = w\psi_n$

$\phi \equiv c_1\psi_1 + c_2\psi_2 + \cdots + c_n\psi_n$

3-D many-particle Schrödinger equation

The 1-D Harmonic Oscillator

$$T = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 \quad , V = \frac{1}{2}kx^2 = 2\pi^2\nu^2 m x^2 \quad , \qquad \nu = \frac{1}{2\pi}\left(\frac{k}{m}\right)^{1/2}$$



Quantum-Mechanical Treatment

$$\frac{d^2\psi}{dx^2} + (2mE\hbar^{-2} - \alpha^2 x^2)\psi = 0 \qquad \alpha \equiv \frac{2\pi\nu m}{\hbar}$$

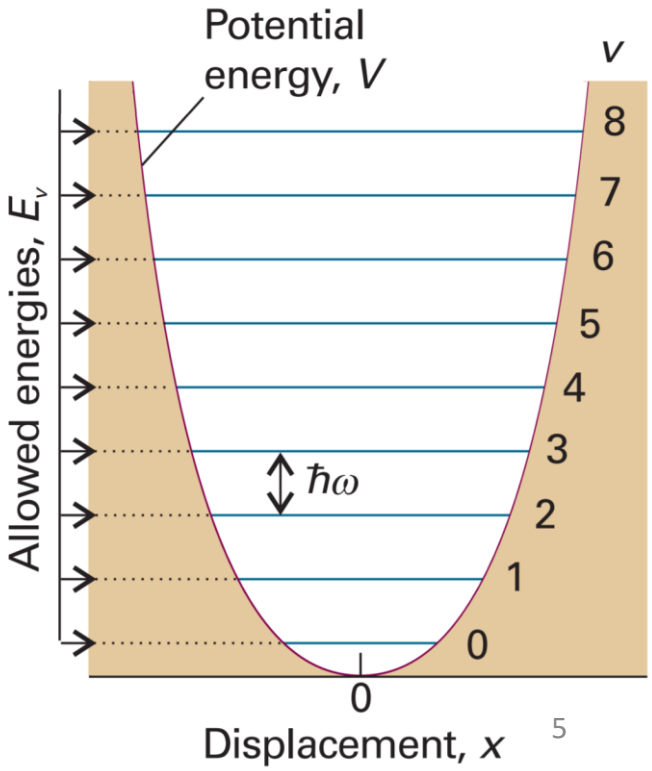
$$\psi = Ae^{-\alpha x^2/2}\sum_{l=0}^{\infty} c_{2l+1}x^{2l+1} + Be^{-\alpha x^2/2}\sum_{l=0}^{\infty} c_{2l}x^{2l}$$

$$E_\upsilon = (\upsilon + \frac{1}{2})h\nu, \quad \upsilon = 0,1,2,\cdots$$

$$E_\upsilon = (\upsilon + \frac{1}{2})\hbar\omega, \quad \omega = \left(\frac{k}{m}\right)^{1/2}$$

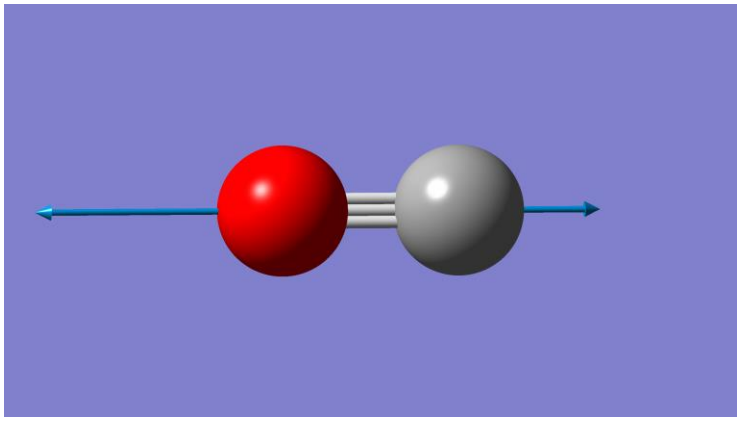
$$\longrightarrow \quad E_0 = \frac{1}{2}\hbar\omega$$

Zero-point energy



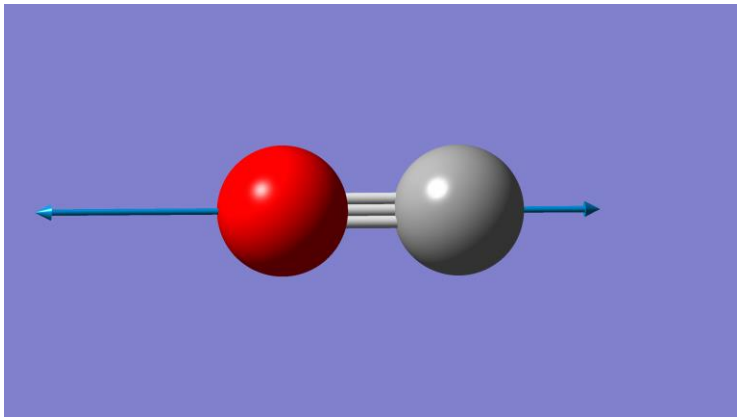
e.g.

The strongest infrared band of $^{12}C^{16}O$ occurs at 2143 cm^{-1}. (a) Find the force constant of $^{12}C^{16}O$; (b) Find the approximate zero-point energy of $^{12}C^{16}O$.



e.g.

The strongest infrared band of $^{12}C^{16}O$ occurs at 2143 cm^{-1}. (a) Find the force constant of $^{12}C^{16}O$; (b) Find the approximate zero-point energy of $^{12}C^{16}O$.



Frequency:

$$v = \tilde{\nu} c$$

Speed of light

Wavenumber

Reduced mass:

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Force Constant:

$$k = 4\pi^2 v^2 \mu$$

Some notations for the integrals

Bracket notation by Dirac.

$$\int f_m^* \hat{A} f_n d\tau \equiv \langle f_m | \hat{A} | f_n \rangle \equiv (f_m | \hat{A} | f_n) \equiv \langle m | \hat{A} | n \rangle \equiv \mathrm{A}_{mn}$$

Bra **Ket**

Matrix-element notation.

Matrix element of operator $\hat{A}$

$$\int f_m^* f_n d\tau \equiv \langle f_m | f_n \rangle \equiv (f_m , f_n) \equiv \langle m | n \rangle$$

$$\langle m | n \rangle^* = \langle n | m \rangle$$

e.g.

$\langle m \mid n \rangle$	**Overlap matrix element**
$\langle m \mid H \mid n \rangle$	**Hamiltonian matrix element**

Five minutes break!



Measurement of a quantum-mechanical system

How to get information about the property *B by Q.M.*?

Postulate 1. The state of a system is described by a function **Ψ, which contains all the information** that can be determined about the system.

Postulate 2. Every physically observable **property ↔ a linear Hermitian operator**.

Ψ **?** → 

A measurement of a property

Postulate 1. The state of a system is described by a function **Ψ, which contains all the information** that can be determined about the system.

Postulate 2. Every physically observable **property ↔ a linear Hermitian operator**.

Postulate 3. The only possible values that can result from measurements of the physically observed property B are **the eigenvalues b_i** in the equation $\hat{B}g_i = b_i g_i$, where $\hat{B}$ is the operator corresponding to the property B. **The eigenfunctions g_i are required to be well-behaved.**

Well-behaved function: finite everywhere; quadratic integrable; "acceptable".

Acceptable wavefunction: continuous single-valued; square-integrable.

e.g.

$$\hat{H}\psi = E\psi$$

---

e.g.

$$\left.\begin{aligned} \hat{p}_x g &= kg \\ \frac{\hbar}{i}\frac{dg}{dx} &= kg \end{aligned}\right\} \quad g = Ae^{ikx/\hbar}$$

To keep g finite for large $|x|$:
Eigenvalues k of $\hat{p}_x$ are the real numbers $-\infty < k < \infty$

How does Ψ give information about the property *B*?

If Ψ is an eigenfunction of $\hat{B}$ with eigenvalue b_k, then a measurement of *B* is certain to yield the value b_k.

Note: The state function Ψ need not be an eigenfunction of the operator $\hat{B}$.

How does Ψ give information about the property *B*?

Note: **The state function Ψ need not be an eigenfunction g_i of the operator $\hat{B}$.**

e.g. **A particle in 1-D box**

$$\Psi(x,t) = e^{-iEt/\hbar}\left(\frac{2}{l}\right)^{1/2}\sin\left(\frac{n\pi x}{l}\right)$$

is **not** an eigenfunction of $\hat{p}_x$

$$\hat{p}_x\Psi = \frac{\hbar}{i}\frac{\partial}{\partial x}e^{-iEt/\hbar}\left(\frac{2}{l}\right)^{1/2}\sin\left(\frac{n\pi x}{l}\right) = \frac{n\pi\hbar}{il}e^{-iEt/\hbar}\left(\frac{2}{l}\right)^{1/2}\cos\left(\frac{n\pi x}{l}\right)$$

$\neq$ constant*Ψ

Measure B $\Rightarrow$

- $\hat{B}\psi = b_k\psi \quad \Rightarrow b_k$
- $\hat{B}\psi \neq b_k\psi \quad \Rightarrow$ **Get one of b_i s**

Measure B $\Rightarrow$

$\hat{B}\psi = b_k\psi \quad \Rightarrow b_k$

$\hat{B}\psi \neq b_k\psi \quad \Rightarrow$ **Get one of b_i s,**

but which one? $\Rightarrow$ **Probability**

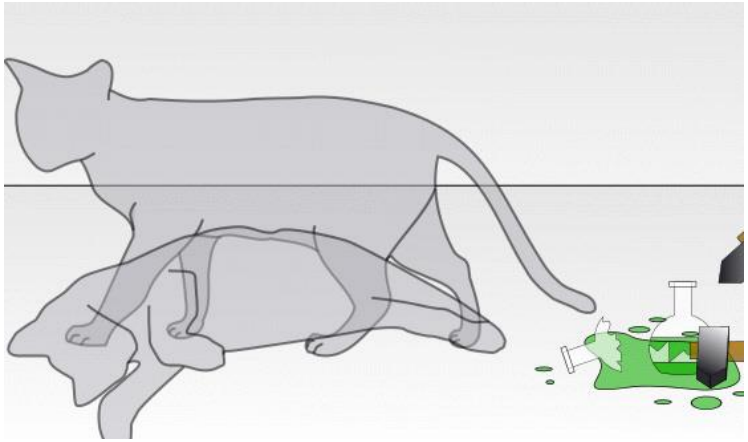
?

How to use Ψ to calculate the probabilities for each possible result of a measurement of B**?**

Postulate 4. If $\hat{B}$ is any linear Hermitian operator that represents a physically observable property, then the eigenfunctions g_i of $\hat{B}$ form a complete set.

The superposition of states (态的叠加)

$$\Psi = \sum_i c_i g_i$$



$$|\Psi\rangle = \frac{|\text{alive}\rangle + |\text{dead}\rangle}{\sqrt{2}}$$

Average value (平均值): the expectation values of an observable

Postulate 5. If $\Psi(q,t)$ is the normalized state function of a system at time t, then the average value of a physical observable B at time t is

$$\langle B \rangle = \int \Psi^* \hat{B} \Psi d\tau$$

For a stationary state: $\langle B \rangle = \int \psi^* \hat{B} \psi d\tau$

If Ψ is an normalized eigenfunction of $\hat{B}$, i.e. $\hat{B}\Psi = k\Psi$,

$$\langle B \rangle = \int \Psi^* \hat{B} \Psi d\tau = \int \Psi^* k \Psi d\tau = k \int \Psi^* \Psi d\tau = k$$

$$\hat{B} g_i(q) = b_i g_i(q) \qquad g_i \text{ are orthonormal}$$

$$\Psi(q,t) = \sum_i c_i(t) g_i(q) \qquad \int \Psi^* \Psi \, d\tau = 1$$

$$\langle B \rangle = \int \Psi^*(q,t) \hat{B} \Psi(q,t) \, d\tau$$

$$\langle B \rangle = \int \sum_i c_i^* g_i^* \hat{B} \sum_k c_k g_k \, d\tau = \sum_i \sum_k c_i^* c_k \int g_i^* \hat{B} g_k \, d\tau$$

$$\langle B \rangle = \sum_i \sum_k c_i^* c_k b_k \int g_i^* g_k \, d\tau = \sum_i \sum_k c_i^* c_k b_k \delta_{ik}$$

$$\rightarrow \quad \langle B \rangle = \sum_i |c_i|^2 b_i \quad \rightarrow \quad \langle B \rangle = \sum_{b_i} P(b_i) b_i$$

The probability of getting b_i

Five minutes break!



e.g.

An operator A has eigenfunctions $f_1, f_2, \ldots, f_n$ with corresponding eigenvalues $a_1, a_2, \ldots, a_n$. The state of a system is described by a normalized wavefunction given by

$$\psi = \frac{1}{3} f_2 + \left(\frac{7}{9}\right)^{1/2} f_4 - \frac{1}{3} i f_5$$

What will be the outcome of measuring the observable A?

e.g.

An operator A has eigenfunctions $f_1, f_2, \ldots, f_n$ with corresponding eigenvalues $a_1, a_2, \ldots, a_n$. The state of a system is described by a normalized wavefunction given by

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What will be the outcome of measuring the observable A?

Answer:

$$\langle A \rangle = \frac{1}{9} a_2 + \frac{7}{9} a_4 + \frac{1}{9} a_5$$

$$\langle B \rangle = \sum_i |c_i|^2 b_i$$

e.g. The energy of a particle of mass *m* in a one-dimensional box of length *l* is measured. What are the possible values that can result from the measurement if at the time the measurement begins, the particle's state function is

$$\Psi = \left(2/l\right)^{1/2} \sin\left(3\pi x/l\right) \qquad for \qquad 0 \le x \le l$$

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$$\Psi = (2/l)^{1/2} \sin(3\pi x/l) \qquad for \qquad 0 \le x \le l$$

$$\psi_n(x) = \left(\frac{2}{L}\right)^{1/2} \sin(n\pi x/L)$$

$$E_n = \frac{n^2 h^2}{8mL^2} \qquad n = 1, 2, \ldots$$

e.g.

$\langle x \rangle$ and $\langle p_x \rangle$ for the ground stationary state of a particle in a 3-D box?

$$\langle x \rangle = \frac{a}{2}$$

$$\langle p_x \rangle = 0$$

$$\psi(x,y,z) = \left(\frac{8}{abc}\right)^{1/2} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

$\langle x \rangle$ for the ground stationary state of a particle in a 3-D box?

$$\psi(x,y,z) = f(x)g(y)h(z)$$

$$\langle x \rangle = \int \psi^* \hat{x} \psi \, d\tau = \int_0^c \int_0^b \int_0^a f^* g^* h^* x f g h \, dx \, dy \, dz$$

$$\langle x \rangle = \int_0^a x |f(x)|^2 \, dx \int_0^b |g(y)|^2 \, dy \int_0^c |h(z)|^2 \, dz = \int_0^a x |f(x)|^2 \, dx$$

G.S.: $n_x = 1$, $\quad f(x) = (2/a)^{1/2} \sin(\pi x/a)$

$$\longrightarrow \quad \langle x \rangle = \frac{2}{a} \int_0^a x \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{a}{2}$$

$$\psi(x,y,z) = \left(\frac{8}{abc}\right)^{1/2} \sin(\frac{n_x \pi x}{a}) \sin(\frac{n_y \pi y}{b}) \sin(\frac{n_z \pi z}{c})$$

$\langle p_x \rangle$ for the ground stationary state of a particle in a 3-D box?

$$\psi(x,y,z) = f(x)g(y)h(z)$$

$$\langle p_x \rangle = \int \psi^* \hat{p}_x \psi \, d\tau = \int_0^c \int_0^b \int_0^a f^* g^* h^* \frac{\hbar}{i} \frac{\partial}{\partial x} [f(x)g(y)h(z)] \, dx \, dy \, dz$$

$$\langle p_x \rangle = \frac{\hbar}{i} \int_0^a f^*(x) f'(x) \, dx \int_0^b |g(y)|^2 \, dy \int_0^c |h(z)|^2 \, dz$$

$$\langle p_x \rangle = \frac{\hbar}{i} \int_0^a f(x) f'(x) dx = \frac{\hbar}{2i} f^2(x) \Big|_0^a = 0$$

$$f(0) = 0 \text{ and } f(a) = 0$$

The properties of oscillators

$$\langle B \rangle = \int \psi^* \hat{B} \psi d\tau$$

The mean displacement of an oscillator: $\langle x \rangle = 0$

The mean square displacement of an oscillator: $\langle x^2 \rangle = \left(\upsilon + \frac{1}{2} \right) \frac{\hbar}{(mk)^{1/2}}$

The mean potential energy of an oscillator:

$$\langle V \rangle = \left\langle \frac{1}{2} k x^2 \right\rangle = \frac{1}{2} \left(\upsilon + \frac{1}{2} \right) \hbar \left(\frac{k}{m} \right)^{1/2} = \frac{1}{2} \left(\upsilon + \frac{1}{2} \right) \hbar \omega = \frac{1}{2} E_\upsilon$$

The mean kinetic energy of an oscillator: $\langle E_K \rangle = \frac{1}{2} E_\upsilon$

Postulates of quantum mechanics

Postulate 1. The state of a system is described by a function **Ψ** of the coordinates and the time. This function, called the state function or **wave function**, contains all the information that can be determined about the system.

Postulate 2. To every physically observable property there corresponds a linear **Hermitian operator**. To find this operator, write down the classical-mechanical expression for the observable in terms of Cartesian coordinates and corresponding linear-momentum components, and then replace each coordinate x by the operator $x\bullet$ and each momentum component p_x by the operator $-i\hbar\partial/\partial x$

Postulate 3. The only possible values that can result from measurements of the physically observed property B are the eigenvalues b_i in the equation $\hat{B}g_i = b_i g_i$, where $\hat{B}$ is the operator corresponding to the property B. **The eigenfunctions g_i are required to be well-behaved**.

Postulate 4. If $\hat{B}$ is any linear Hermitian operator that represents a physically observable property, then the **eigenfunctions g_i of $\hat{B}$ form a complete set**.

Postulate 5. If $\Psi(q,t)$ is the normalized state function of a system at time t, then the average value of a physical observable B at time t is $\langle B \rangle = \int \Psi^* \hat{B} \Psi d\tau$

Postulate 6. The time development of the state of an undisturbed quantum-mechanical system is given by the Schrödinger time-dependent equation.

Three minutes break!



Chapter 1

Quantum Theory

Part III

Contents

- The Variation Theorem
- Hydrogenic Atoms

The Variation Theorem.

Given a system whose Hamiltonian operator $\hat{H}$ is time independent and whose lowest-energy eigenvalue is E_1, if **ϕ is any normalized well-behaved function** of the coordinates of the system's particles that satisfies the boundary conditions of the problem, then

$$\int \phi^* \hat{H} \phi d\tau \geq E_1$$

Expanding ϕ in terms of ***the complete, orthonormal set*** of eigenfunction of $\hat{H}$

$$\hat{H}\psi_k = E\psi_k$$

$$\phi = \sum_k a_k \psi_k$$

$$\int \phi^* \hat{H} \phi d\tau = \int \sum_k a_k^* \psi_k^* \hat{H} \sum_j a_j \psi_j d\tau$$

$$= \sum_k \sum_j a_k^* a_j E_j \delta_{kj}$$

since $E_k \geq E_1$ ⟶ $= \sum_k |a_k|^2 E_k \geq E_1$

If ϕ **is not normalized, the variation theorem:**

$$\frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} \geq E_1$$

If ϕ **is not normalized, the variation theorem:**

$$\frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} \geq E_1$$

Normalization $\phi \rightarrow N\phi$

$$\int (N\phi)^* (N\phi) d\tau = |N|^2 \int \phi^* \phi d\tau = 1 .$$

$$|N|^2 = 1 \Big/ \int \phi^* \phi d\tau$$

$$\rightarrow \quad |N|^2 \int \phi^* H \phi d\tau \geq E_1 \quad \rightarrow \quad \frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} \geq E_1$$

The variation method

If ϕ is normalized $\int \phi^* \hat{H} \phi d\tau \geq E_1$

To arrive at good approximation to the ground state energy E_1, we try many trial variation functions and look for the one that gives the lowest value of the variational integral.

ϕ : **trial variational function**

$E = \int \phi^* \hat{H} \phi d\tau$ **variational integral**

The lower the value of the variational integral, the better approximation we have to E_1.

The variation method

e.g. Predict the ground state energy of a particle in a 1-D Box by the variation method.

The wave function is zero outside the box;
The boundary conditions are:
$\psi = 0$ at $x = 0$ and at $x = l$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}$$

A trial variational function $\phi = x(l-x)$

$$\frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} = \frac{-\frac{\hbar^2}{2m}\int x(l-x)\frac{d^2}{dx^2}[x(l-x)]dx}{\int x^2(l-x)^2 dx}$$

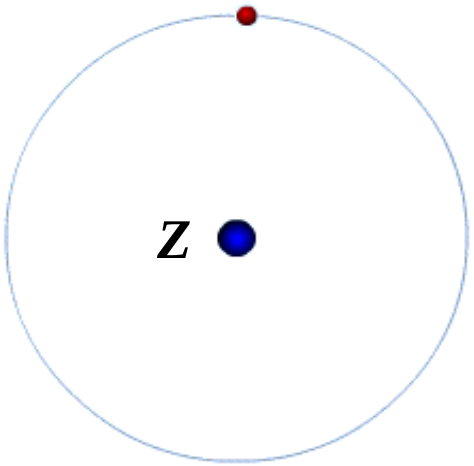
$$= \frac{\frac{\hbar^2}{2m}\int (2xl - 2x^2)dx}{\int (x^4 + l^2x^2 - 2lx^3)dx} = \frac{\frac{\hbar^2 l^3}{6m}}{\frac{l^5}{30}} = \frac{5h^2}{4\pi^2 ml^2} = 0.1266515\frac{h^2}{ml^2}$$

$$E_1 = \frac{h^2}{8ml^2} = 0.125\frac{h^2}{ml^2}$$

Two minutes break!



Hydrogenic (hydrogenlike) Atoms 类氢原子



A hydrogenic atom:
a one-electron atom or one-electron ion of general atomic number Z;
e.g., H, He^{+}, Li^{2+},O^{7+}, U^{91+}…

- Their Schrödinger equations can be solved exactly;
- Provide a set of concepts to describe the structures of many-electron atoms and molecules.

The Hamiltonian for a hydrogenic atom:

$$\hat{H} = \hat{E}_{\mathrm{K,electron}} + \hat{E}_{\mathrm{K,nucleus}} + \hat{V}$$

$$= -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_N}\nabla_N^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$$

Convert to center-of-mass coordinates

Separate to two equations

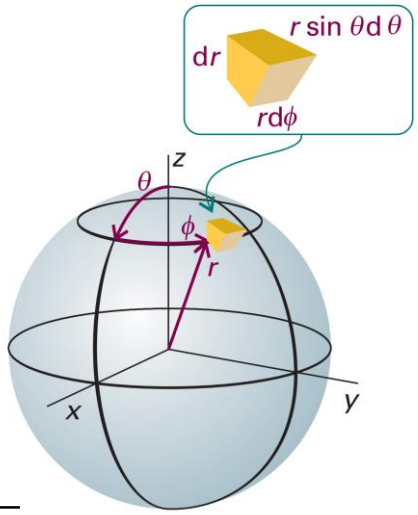
- **The motion of the atom as a whole**
- **The motion of the electron relative to the nucleus**

1-*e* Schrödinger equation in spherical polar coordinates:

$$\left(\frac{-\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)\ ,$$

$$\frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_N}$$

The reduced mass



$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

Hydrogenic Atoms

1-*e* Schrödinger equation in spherical polar coordinates:

$$\left(\frac{-\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \underbrace{\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}}_{\frac{1}{r^2}\Lambda^2}$$

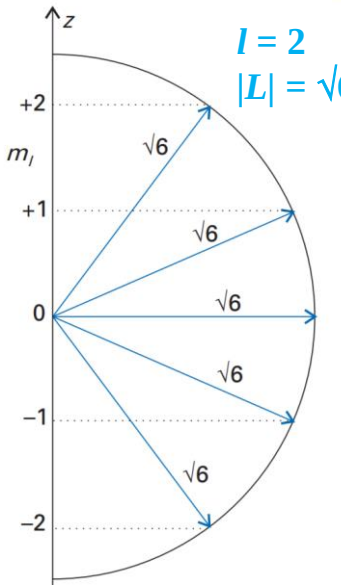
$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

$$\rightarrow \begin{cases} \Lambda^2 Y = -l(l+1)Y & \text{Angular part} \\ -\dfrac{\hbar^2}{2\mu}\dfrac{\mathrm{d}^2 u}{\mathrm{d}r^2} + V_{\text{eff}}u = Eu \qquad u = rR & \text{Radial part} \end{cases}$$

$$V_{\text{eff}} = -\frac{Ze^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2\mu r^2}$$

Angular wave equation

$$\Lambda^2 Y = -l(l+1)Y$$



$l = 2$
$|L| = \sqrt{6}\hbar$

l, m specify the angular momentum of the electron around the nucleus.

l,	m,	$Y_{l,m}(\theta,\phi)$
0	0	$\left(\frac{1}{4\pi}\right)^{1/2}$
1	0	$\left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$
	±1	$\mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta\, e^{\pm i\phi}$
2	0	$\left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta-1)$
	±1	$\mp\left(\frac{15}{8\pi}\right)^{1/2}\cos\theta\sin\theta\, e^{\pm i\phi}$
	±2	$\left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta\, e^{\pm 2i\phi}$

Orbital with **orbital angular momentum quantum number** ***l*** has an angular momentum of magnitude $\{l(l+1)\}^{1/2}\hbar$, with $l = 0, 1, 2, \ldots, n-1$.

Orbital with **magnetic quantum number** ***m*** has a *z*-component of angular momentum $m\hbar$, with $m = 0, \pm 1, \pm 2, \ldots, \pm l$.

Hydrogenic Atoms

Radial wave equation

$$-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2}+V_{\text{eff}}u=Eu \qquad u=rR$$

$$E_n=-\frac{Z^2\mu e^4}{32\pi^2\varepsilon_0^2\hbar^2n^2}$$

Principal quantum number ***n***, determines the energy, ***n* = 1, 2, 3, ...**

Orbital	n	l	$R_{n,l}$
$1s$	1	0	$2\left(\frac{Z}{a}\right)^{3/2}e^{-\rho/2}$
$2s$	2	0	$\frac{1}{8^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(2-\rho)e^{-\rho/2}$
$2p$	2	1	$\frac{1}{24^{1/2}}\left(\frac{Z}{a}\right)^{3/2}\rho e^{-\rho/2}$
$3s$	3	0	$\frac{1}{243^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(6-6\rho+\rho^2)e^{-\rho/2}$
$3p$	3	1	$\frac{1}{486^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(4-\rho)\rho e^{-\rho/2}$
$3d$	3	2	$\frac{1}{2430^{1/2}}\left(\frac{Z}{a}\right)^{3/2}\rho^2e^{-\rho/2}$

$$\rho=\frac{2Zr}{na}$$

Bohr radius

$$a=\frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}=0.529 \quad \text{Å}$$

Atomic orbital and energy

An atomic orbital: a one-electron wavefunction for an electron in an atom, defined by **the 3 quantum numbers, *n*, *l*, *m*.**

Principal quantum number（主量子数） ***n*,** → the energy of the orbital, ***n* = 1, 2, 3, ...**

Angular momentum quantum numbers (角量子数) *l*, *m* specify the angular momentum of the electron around the nucleus.

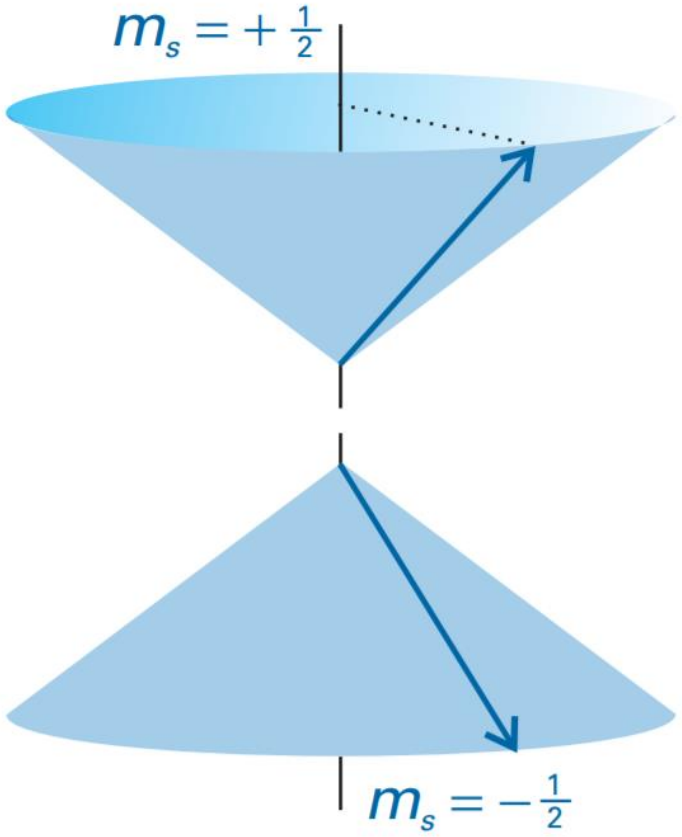
Angular momentum of magnitude $\{l(l+1)\}^{1/2}\hbar$, with $l = 0,1, 2, \ldots, n-1$.

z-component of angular momentum $m\hbar$, with $m = 0, \pm1, \pm 2, \ldots, \pm l$.

Specify a state of an electron, n, l, m_l, m_s

Intrinsic spin angular momentum quantum numbers (自旋角量子数) of electron, s, m_s

$s = ½$, $m_s = \pm ½$



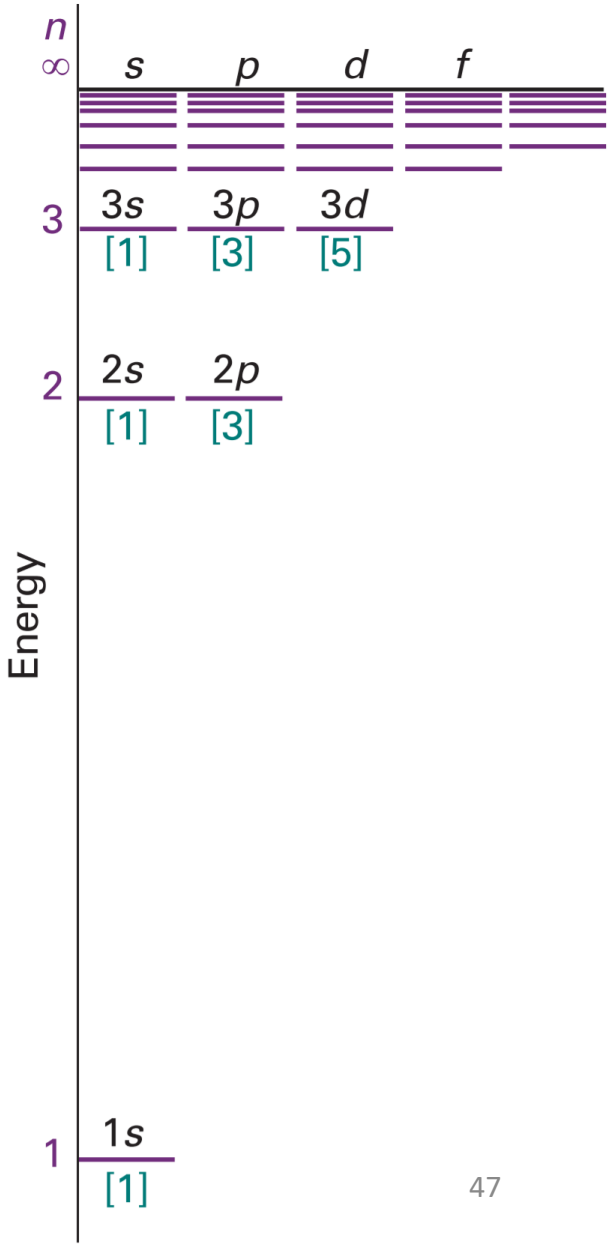
The energy levels

$$E_n = -\left(\frac{Z^2 \mu e^4}{32\pi^2 \varepsilon_0^2 \hbar^2}\right)\frac{1}{n^2} \qquad n = 1, 2, \ldots$$

The higher the value of "*n*" , the greater the energy of the orbital.

The **degeneracy** of the energy level characterized by *n* is $\boldsymbol{n^2}$

The energy levels of a H atom



Shells and Subshells

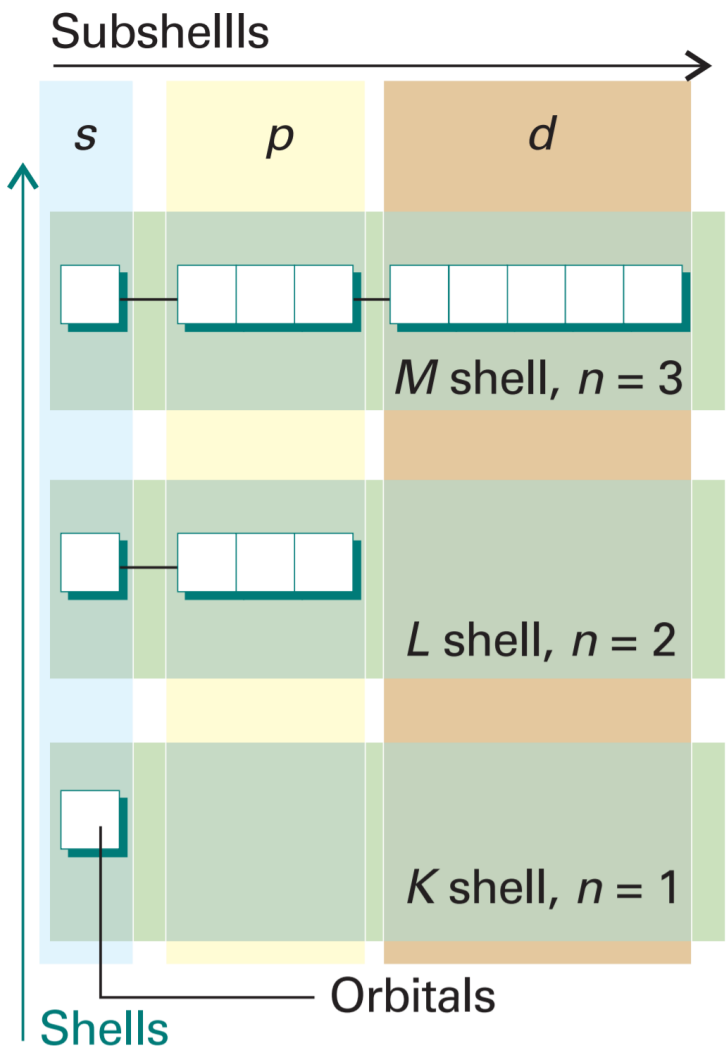
Shell: all the orbitals with the same *n*.

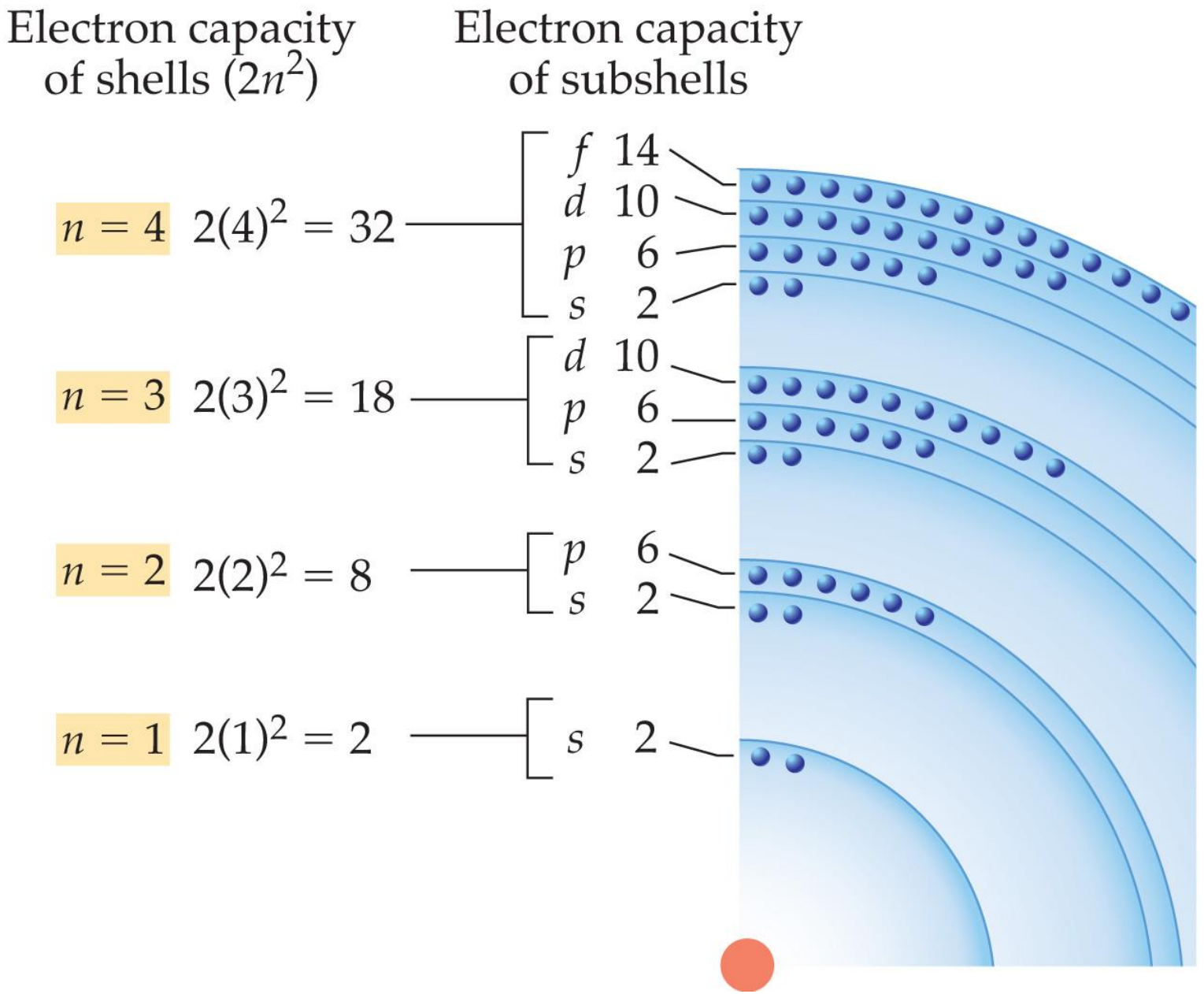
$n =$	1	2	3	4 . . .
	K	L	M	N . . .

Subshell: all the orbitals with the same *n* but different of *l* form a subshell of the given shell.

$l =$	0	1	2	3
	s	p	d	f

4	5	6 . . .
g	h	i . . .



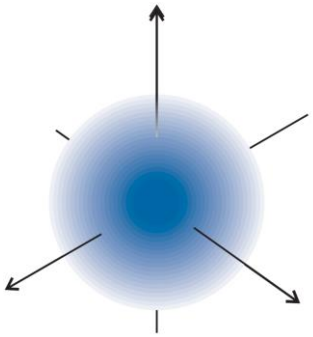

Electron capacity of shells ($2n^2$)
Electron capacity of subshells
$n = 4$ $2(4)^2 = 32$
f 14
d 10
p 6
s 2
$n = 3$ $2(3)^2 = 18$
d 10
p 6
s 2
$n = 2$ $2(2)^2 = 8$
p 6
s 2
$n = 1$ $2(1)^2 = 2$
s 2

s atomic orbital

l,	m,	$Y_{l,m}(\theta,\phi)$
0	0	$\left(\frac{1}{4\pi}\right)^{1/2}$

Spherically symmetrical

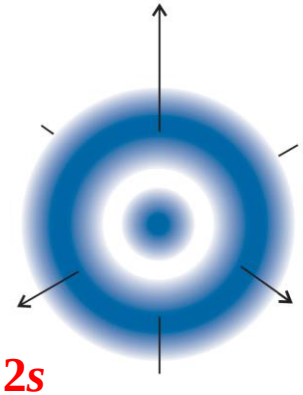
Density of shading



1*s*

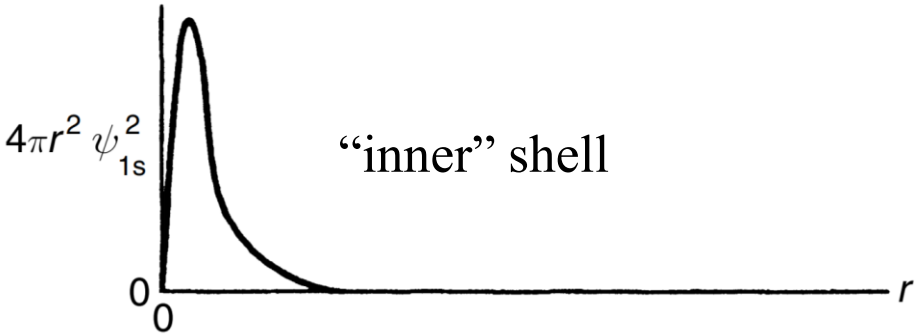
The orbital with $n=1$: $l=0$, $m_l=0$ **1*s***

$n=2$: $l=0$, $m_l=0$ **2*s***

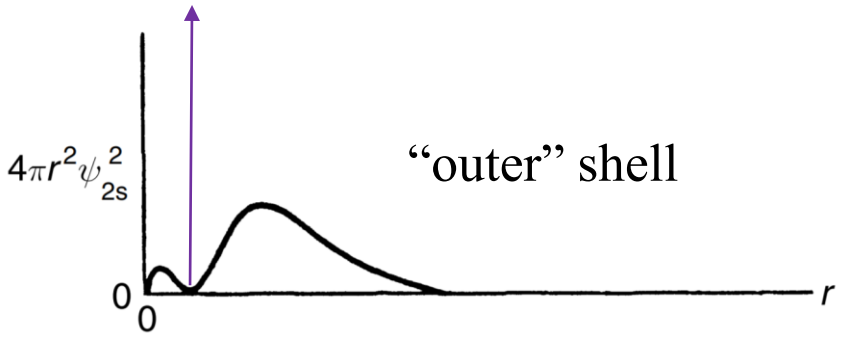


2*s*

Node: a surface on which an electron is not found.



"inner" shell

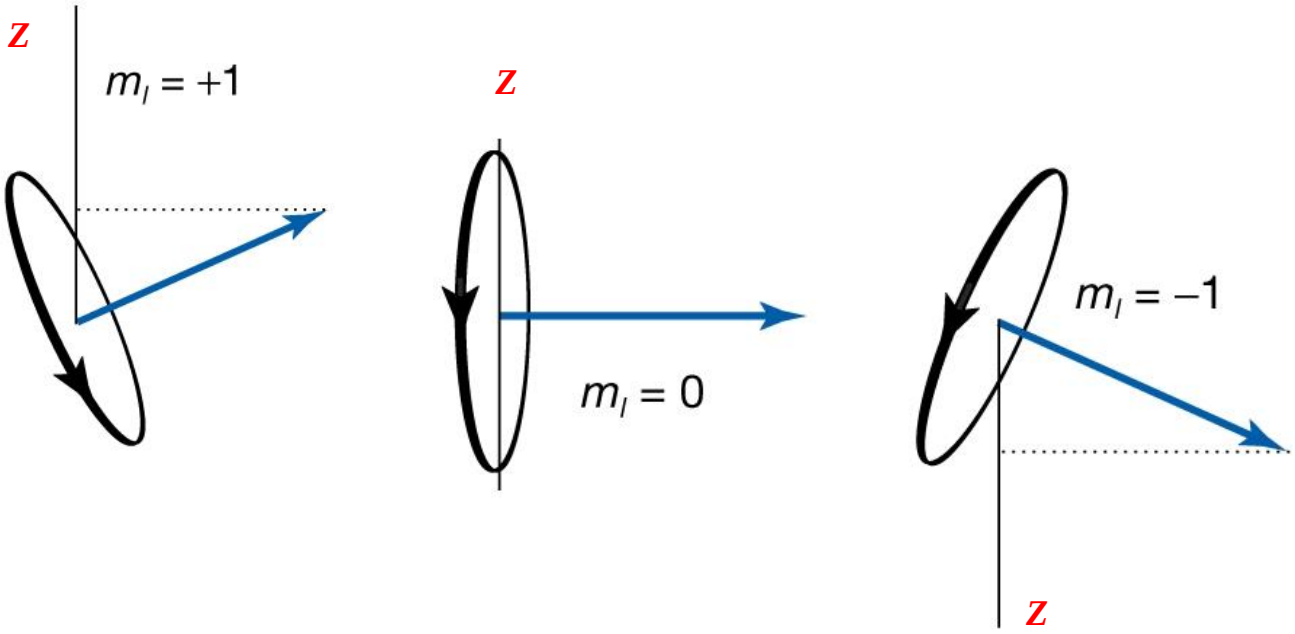


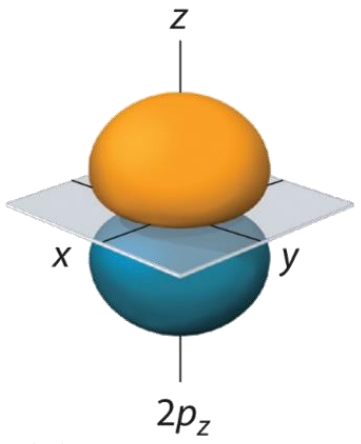
"outer" shell

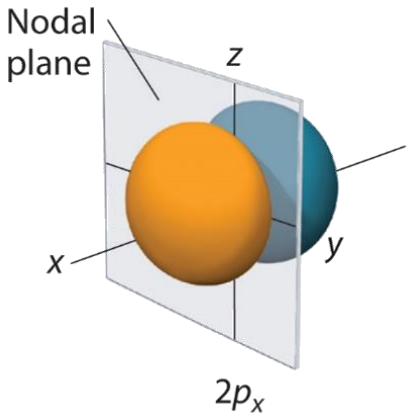
p atomic orbital $l = 1$

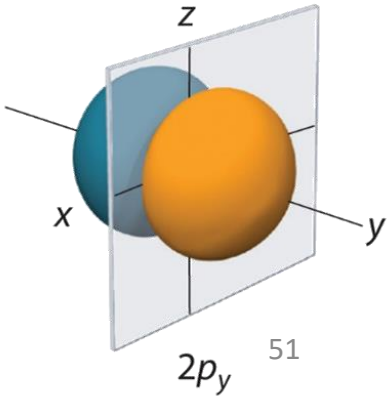
The orbital with $n=2$: $l=1$, $m_l=0$ **$2p_0$ or $2p_z$**

$n=2$: $l=1$, $m_l=\pm1$ **$2p_{\pm1}$ or $2p_x$, $2p_y$**

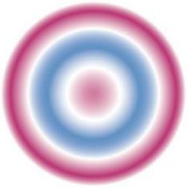




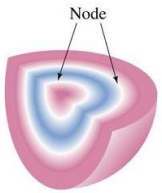


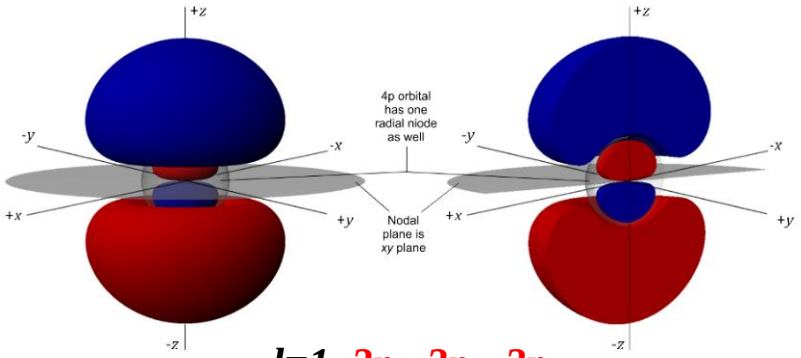


Atomic orbitals with *n*=3



l=0, 3*s*





l=1, $3p_z$, $3p_x$, $3p_y$

l=2, *m*=-2,-1,0,1,2,
$3d_{z^2}$, $3d_{xy}$, $3d_{xz}$, $3d_{yz}$, $3d_{x^2-y^2}$

